

Multivariate Detection & Calibrated Score Fusion

From single metrics to correlated systems

Agenda

- cross-correlation and coherence checks,
- embeddings: PCA and AE for multivariate windows,
- pattern options: Matrix Profile, shapelets, and DTW/template matching,
- fusion formulas, calibration, and reporting,
- guided walkthrough on multivariate panels.

Why this changes the problem

Real systems rarely fail on one metric alone.

Multivariate anomalies often look like:

- broken co-movement,
- lead/lag changes,
- channel-specific deviations,
- or unusual joint shapes across several signals.

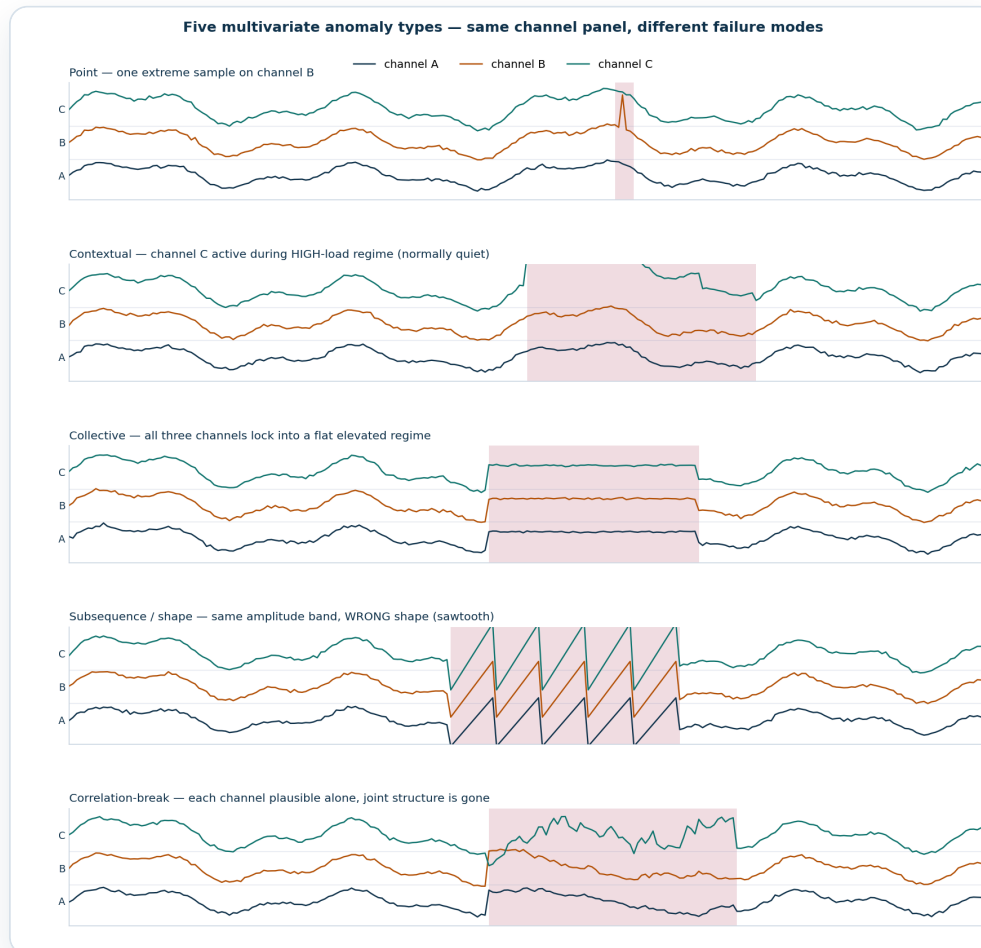
The task is no longer only detection. It is also **triage, calibration, and explanation.**

Five multivariate anomaly types worth distinguishing

- **Point** — one or a few extreme samples.
- **Contextual** — unusual given regime / load / lag context.
- **Collective** — a suspicious run or segment.
- **Subsequence / shape** — the pattern is wrong even if magnitude still looks plausible.
- **Correlation-break** — channels look plausible alone, but the joint structure breaks.

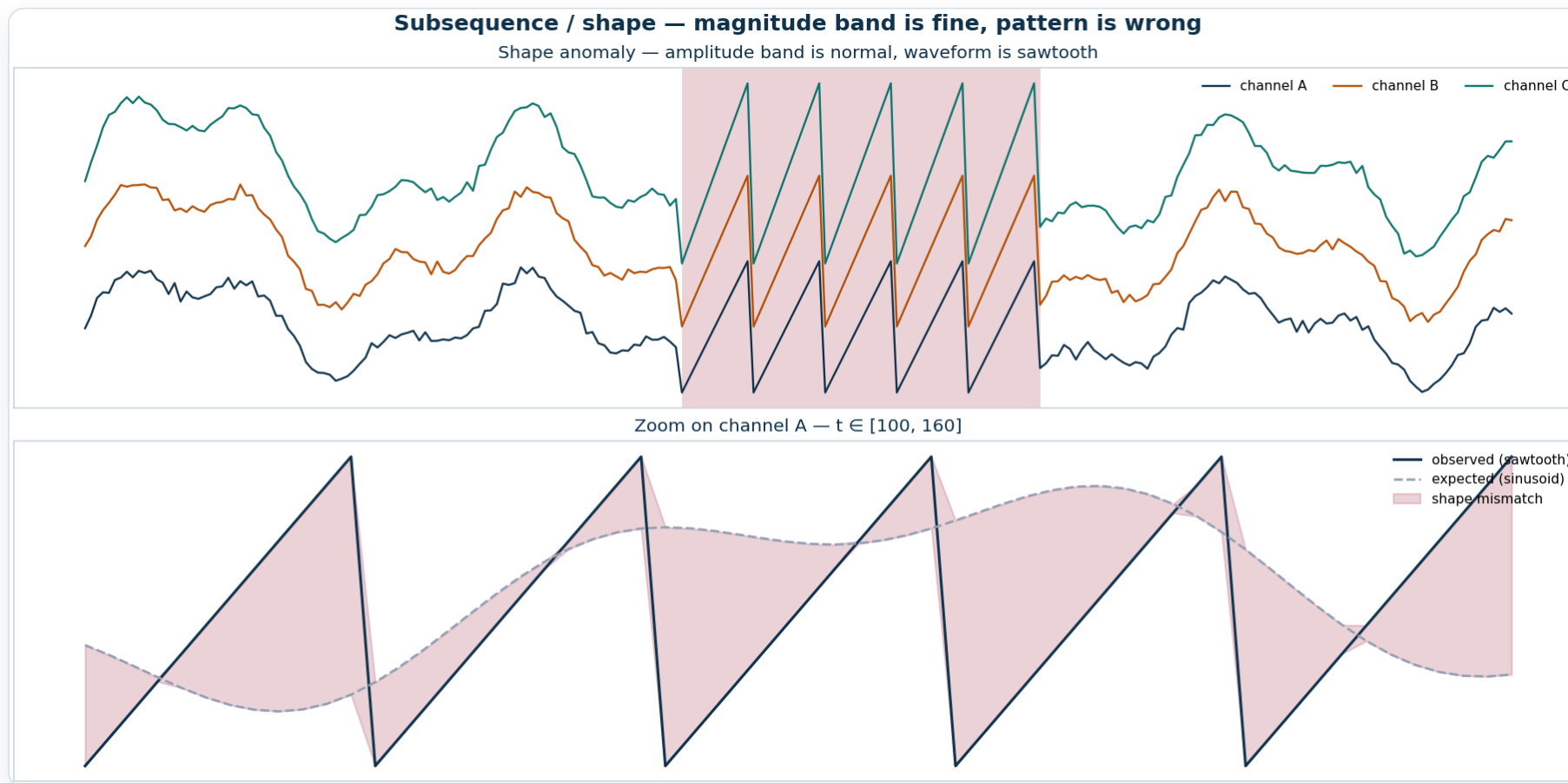
The last two are the ones that **single-channel** detectors miss most consistently. They drive the case for multivariate workflows on the rest of this day.

Five types on the same channel panel



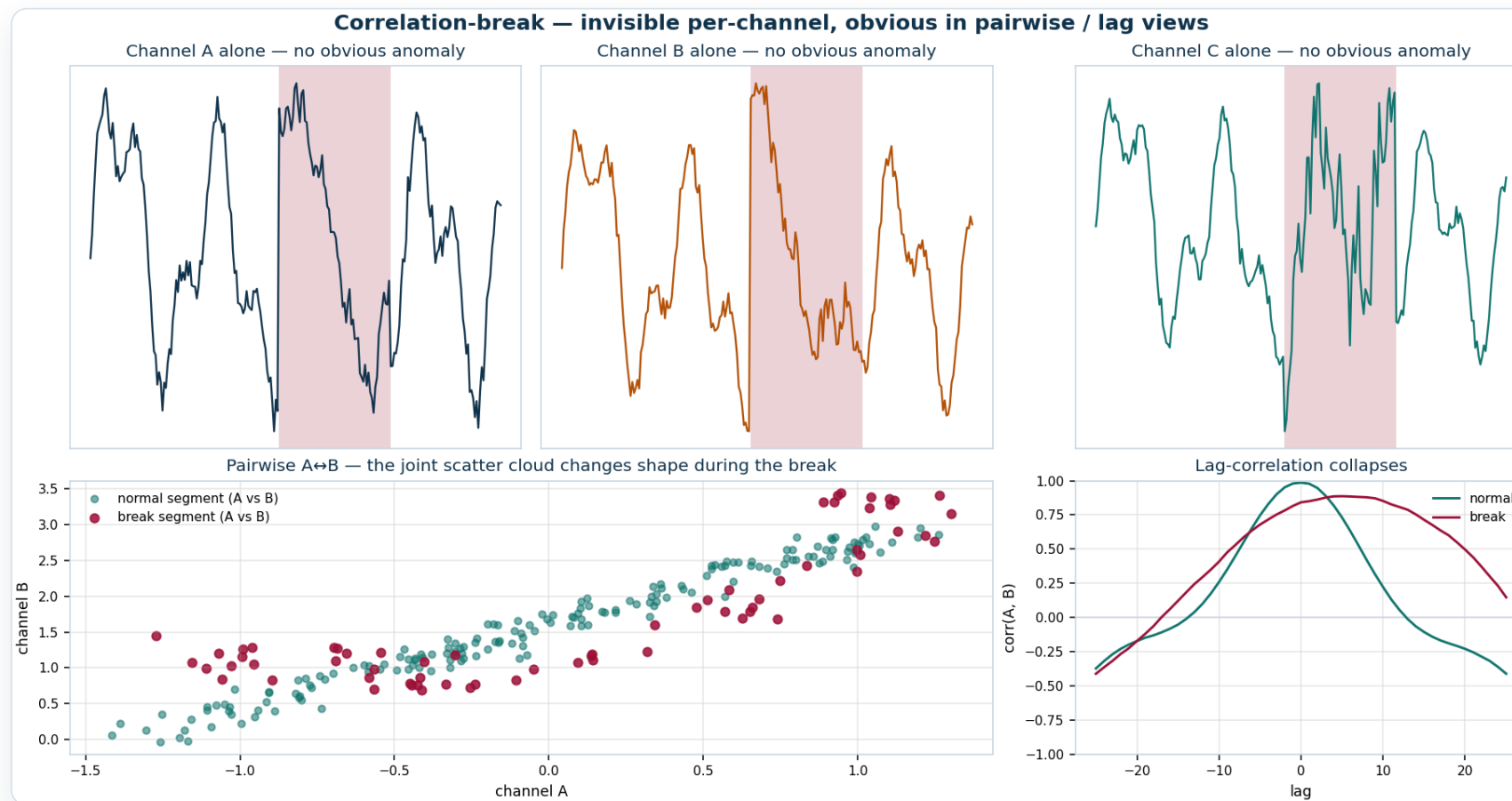
Same three channels, five different failure modes. The shaded region marks the abnormal segment in each row. Notice how the same panel layout reveals very different shapes of failure.

Zoom — Subsequence / shape anomaly



The amplitude band is preserved — a magnitude-only detector (z-score, IF on raw stats) does not fire. The **shape** is wrong: sawtooth where the system normally runs sinusoidal. This is what Matrix Profile, shapelets, and DTW/template-matching are designed to catch.

Zoom — Correlation-break anomaly



Every channel alone passes a univariate review. The break only appears in **pairwise scatter** (the joint A-vs-B cloud changes shape) and in **lag-correlation** (the clean cross-correlation peak collapses). This is the canonical case where univariate alarms stay silent and multivariate methods earn their keep.

Cross-correlation and coherence checks

- **Cross-correlation** helps surface lead/lag relationships in the time domain.
- **Coherence** helps surface shared frequency structure even when the raw traces are noisy.
- Together they help decide:
 - which channels belong in the same review panel,
 - what lag range is worth checking,
 - and whether a relationship break is temporal, spectral, or both.

Professional caveat: common trends can create fake confidence in both correlation and coherence, so detrending or differencing may still be necessary.

Cross-correlation — formula and reading

For two centered channels x and y , inspect lagged alignment with:

$$\rho_{xy}(k) = \text{corr}(x_{t+k}, y_t) = \frac{\sum_t (x_{t+k} - \bar{x})(y_t - \bar{y})}{\sqrt{\sum_t (x_{t+k} - \bar{x})^2} \sqrt{\sum_t (y_t - \bar{y})^2}}$$

- $\max_k |\rho_{xy}(k)|$ answers: **at what offset do these channels align best?**
- In this convention, positive k shifts the first series forward relative to the second.
- Interpretation: useful for review-panel construction and plausible lag ranges; **not** a causal claim.

Power spectra and why we smooth them

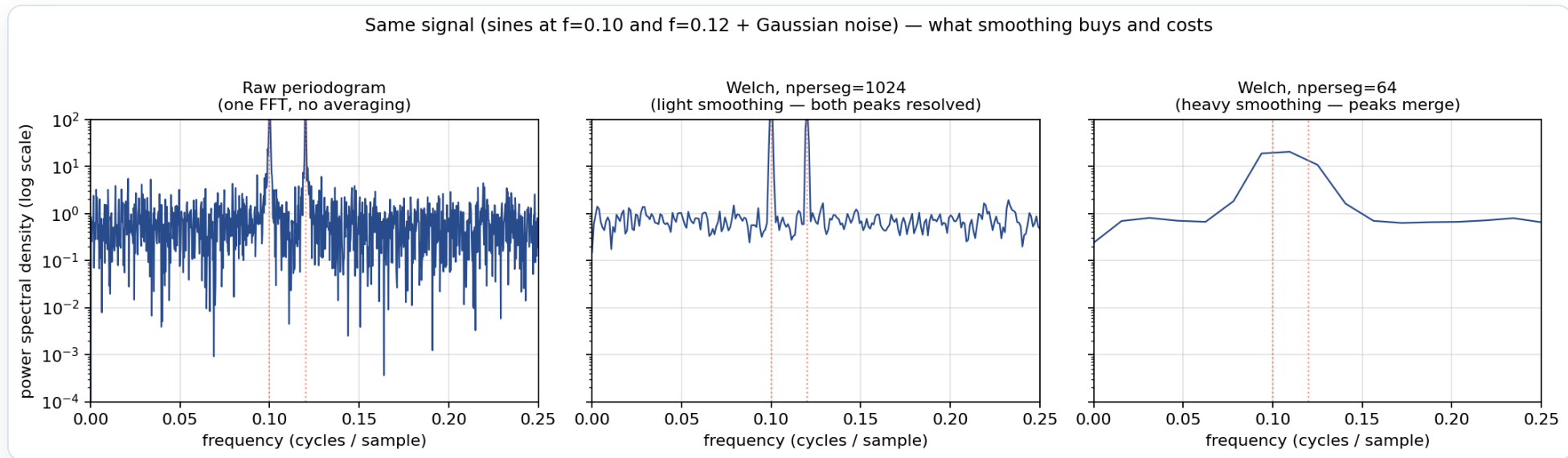
The **power spectrum** $S_{xx}(f) = |X(f)|^2$ shows how the variance of $x(t)$ is distributed across frequencies — a peak at $f = 1/24$ h means a strong daily rhythm.

The **cross-spectrum** $S_{xy}(f) = X(f) \overline{Y(f)}$ measures shared structure between two channels at each frequency.

A single FFT of one finite chunk gives a noisy, jagged estimate — individual bins cannot be trusted. The tilde $\widetilde{S}$ marks the **smoothed** estimate, typically via:

- **Welch's method** — split into overlapping segments, FFT each, average the squared magnitudes.
- **Frequency-band smoothing** — average neighboring bins of one FFT.
- More smoothing $\rightarrow$ lower variance per bin, but coarser frequency resolution. A classical bias/variance trade-off.
- **Why this matters for coherence:** with unsmoothed spectra, $C_{xy}(f)$ collapses algebraically to 1 at every frequency. Smoothing is what turns coherence into a real measurement — “do these channels share structure at f consistently across segments?”

Smoothing — what it buys and costs

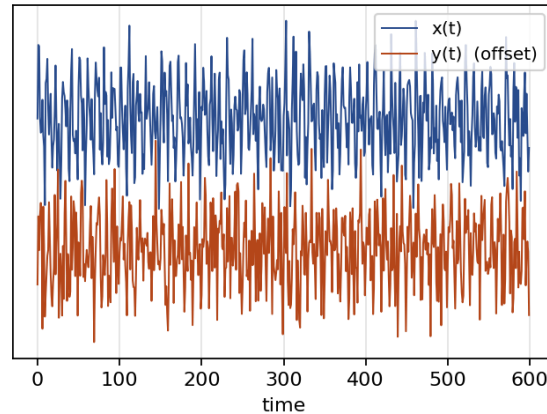


Same signal — two sinusoids at $f = 0.10$ and $f = 0.12$ plus Gaussian noise. **Left:** raw periodogram, peaks buried in variance. **Middle:** light Welch averaging cleans the trace and both peaks survive. **Right:** heavy averaging is smoother still, but the two peaks have **merged** into one bump — that is the resolution cost. The right setting is “smooth enough to trust, sharp enough to separate what you need to separate.”

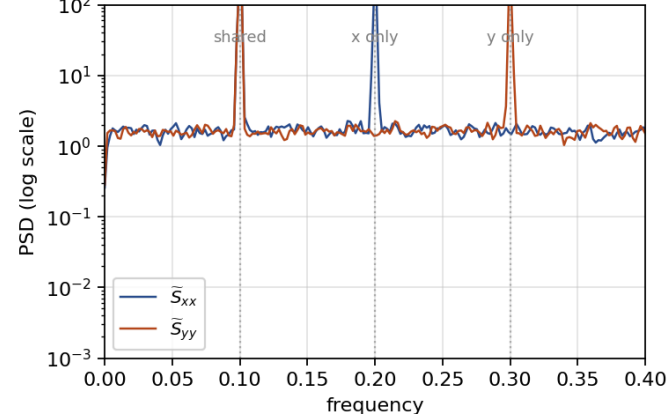
Cross-spectrum — what S_{xy} actually shows

$$S_{xy}(f) = X(f) \overline{Y(f)} \text{ — isolates structure that two channels share at frequency } f$$

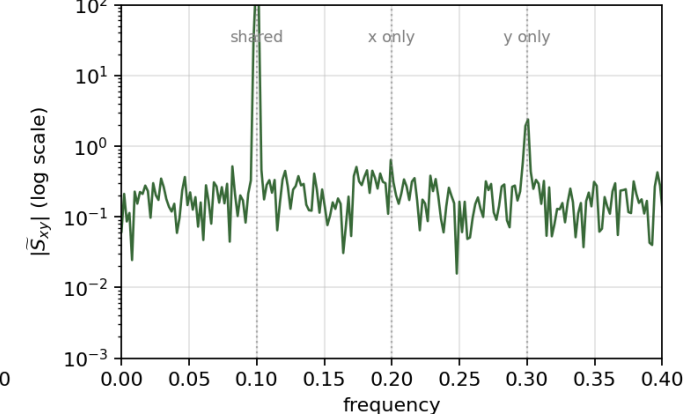
Two noisy channels — shared rhythm not obvious by eye



Auto-spectra: each channel keeps its own peaks



Cross-spectrum: only the shared peak survives



Two channels share one hidden sinusoid at $f = 0.10$; each also carries a private one (x at 0.20, y at 0.30) plus independent noise. **Left:** the raw traces — the shared rhythm is invisible by eye. **Middle:** the smoothed auto-spectra $\bar{S}_{xx}$ and $\bar{S}_{yy}$ each keep their own peaks. **Right:** $|\bar{S}_{xy}|$ keeps only what the two channels actually share — the 0.10 peak survives, the 0.20 and 0.30 peaks collapse into the noise floor because their phases are uncorrelated and average toward zero across segments.

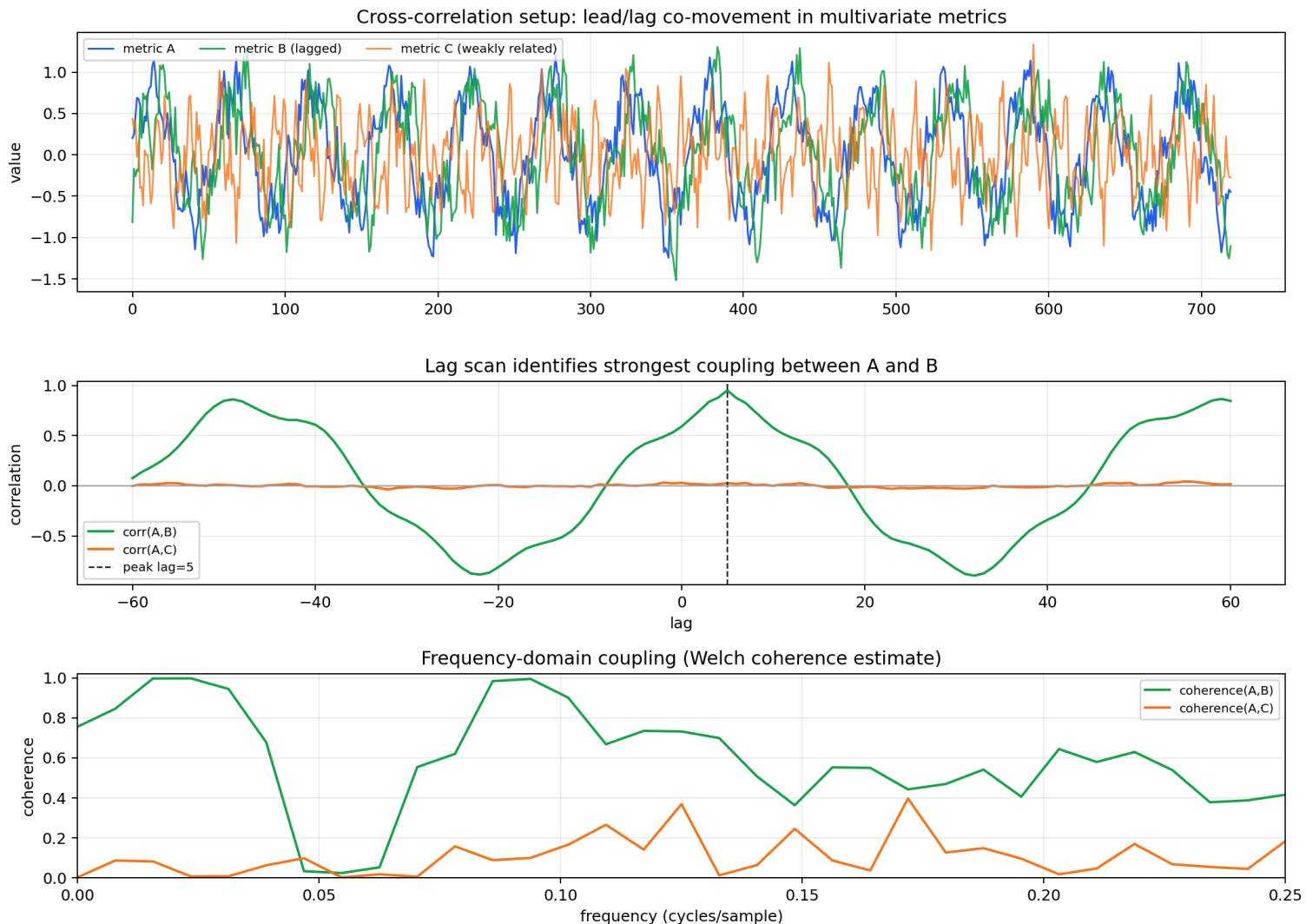
Coherence — formula and reading

Using smoothed spectra:

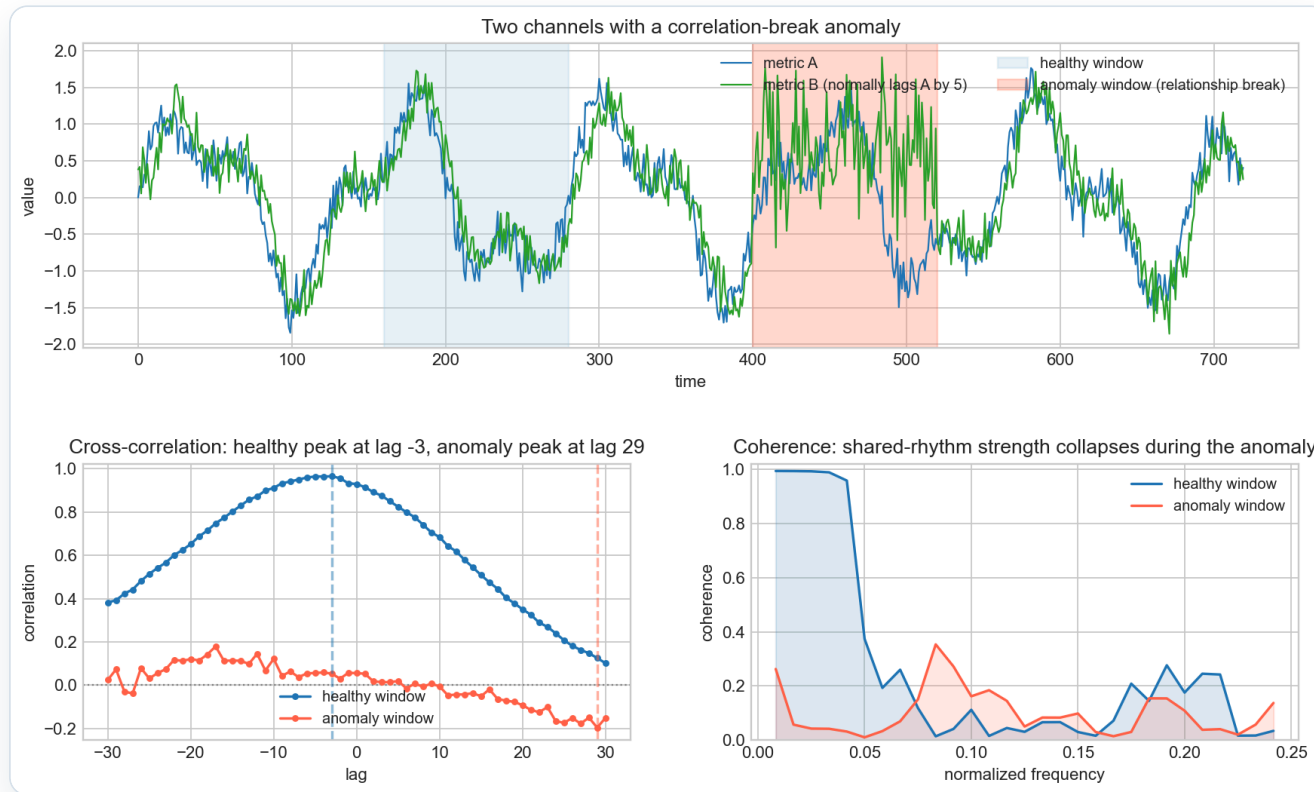
$$C_{xy}(f) = \frac{|\widetilde{S}_{xy}(f)|^2}{\widetilde{S}_{xx}(f)\widetilde{S}_{yy}(f) + \epsilon} \quad \text{with} \quad 0 \leq C_{xy}(f) \leq 1$$

- $\widetilde{S}_{xx}$ and $\widetilde{S}_{yy}$ are smoothed power spectra; $\widetilde{S}_{xy}$ is the smoothed cross-spectrum.
- High coherence means the pair shares structure at that frequency band.
- Interpretation: **same rhythm** can exist even when raw traces look noisy; trend leakage can still fool the measure.

Cross-correlation and coherence — example



Cross-correlation and coherence — with an anomaly



Same pair of channels, compared on a **healthy** window vs an **anomaly** window. Healthy: crisp lag peak near -3 , strong low-frequency coherence. Anomaly: lag profile flattens (no recoverable offset), coherence collapses across the band — the *relationship* breaks even though each channel alone may still look plausible.

Operational anomaly types — and which branch tends to fire

Quick taxonomy anchor:

- **Point** = one or a few extreme samples
- **Contextual** = unusual given regime / load / lag context
- **Collective** = a suspicious run or segment
- **Subsequence / shape** = the pattern is wrong even if magnitude still looks plausible
- **Correlation-break** = channels look plausible alone, but the joint structure breaks

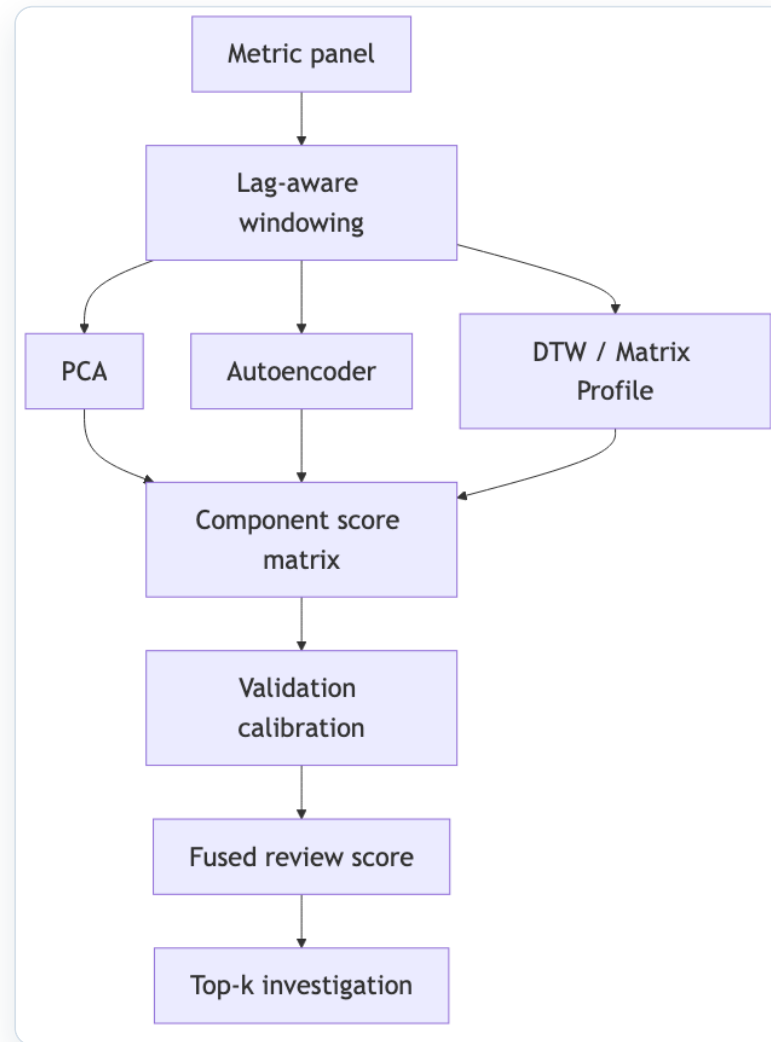
These labels help explain *why* detector branches disagree.

Map anomaly flavor to detector family

Flavor	First branch	Often misses
Point: single-channel spike	AE / residual	joint-space events
Contextual: relationship break	PCA or AE	slow in-manifold drift
Collective: shape mismatch	DTW / template	pure magnitude shifts
Collective: covariance shift	PCA	nonlinear coupling

Use this table when interpreting branch disagreement later in the day.

Multivariate workflow



Windowing and scaling formulas

The training workflow turns a metric panel into fixed-length windows:

$$X_i = \begin{bmatrix} x_i \\ x_{i+1} \\ \dots \\ x_{i+w-1} \end{bmatrix} \in \mathbb{R}^{w \times d}, \quad y_i = \max_{t=i}^{i+w-1} \text{label}_t$$

Then each channel is scaled from healthy training windows:

$$X'_{i,t,j} = \frac{X_{i,t,j} - \mu_j}{\sigma_j + \epsilon}$$

Interpretation: every detector scores the **same normalized window**, so branch disagreement is about model assumptions, not raw units.

Detector roles — PCA & AE

PCA branch

- fast and interpretable,
- strong on linear correlated level shifts,
- good baseline for “something in the joint space moved.”

AE branch

- a **simple dense autoencoder** — the deep baseline against PCA, not a state-of-the-art detector,
- can absorb mild nonlinear cross-feature structure that PCA misses,
- benefits from clean scaling and stable training.

PCA and AE scores — formula and reading

Flatten each window as $v_i = \text{vec}(X'_i)$.

$$s_i^{\text{PCA}} = \frac{1}{wd} \|v_i - \text{PCA}^{-1}(\text{PCA}(v_i))\|_2^2$$

$$s_i^{\text{AE}} = \frac{1}{wd} \|X'_i - g_{\theta}(f_{\theta}(X'_i))\|_F^2$$

- High PCA score: the window moved away from the learned **linear joint subspace**.
- High AE score: the window is hard for the nonlinear reconstruction baseline to reproduce.
- If PCA is strong and AE is weak, the simpler linear baseline may already explain the useful separation.

Detector role — DTW / template

DTW / template branch

- asks: *does this window still look like a healthy shape?*
- complements reconstruction approaches,
- especially helpful when shape matters more than raw magnitude.

DTW template score — formula and reading

For one channel, constrained DTW uses:

$$D_{r,c} = |a_r - b_c| + \min(D_{r-1,c}, D_{r,c-1}, D_{r-1,c-1})$$

The multivariate template score sums channel distances:

$$s_i^{\text{DTW}} = \sum_{j=1}^d \text{DTW}(X'_{i:,j}, T_{:,j})$$

Interpretation: high DTW means the window no longer resembles the selected healthy template after limited time warping. It is template-dependent by design.

Shape-based detectors — the family

- **DTW / template matching** — the default known-reference branch in this module. The companion implementation uses **constrained DTW** (Sakoe–Chiba band); **LB_Keogh** is a scale-up note for longer runs, not part of the core exercise.
- **Multivariate Matrix Profile** — now a runnable companion branch. Score = distance to the nearest non-trivial subsequence; valleys are motifs and peaks are discords. On larger real panels, switch to `stumpy.mstump`.
- **Shapelets / U-shapelets** — discriminative subsequences. *Discussed only — not implemented here.*

Decision rule: template known → **DTW / template**; self-similarity mining / motif-discord review → **Matrix Profile**.

Matrix Profile score — formula and reading

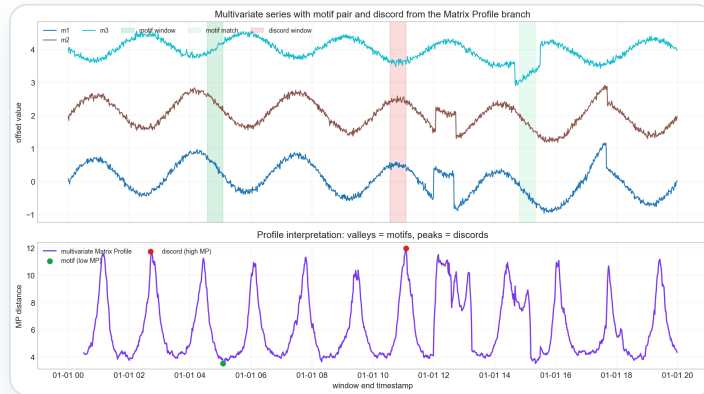
For z-normalized subsequence windows W_i :

$$MP_i = \min_{j: |i-j|>e} \|z(W_i) - z(W_j)\|_2$$

where e is the exclusion zone that prevents trivial near-neighbor matches.

- Low MP_i : this window has a close match elsewhere → **motif**.
- High MP_i : this window has no close non-trivial match → **discord**.
- Interpretation: Matrix Profile is a **self-similarity** branch, not a supervised anomaly classifier.

Matrix Profile review in this pack



- The companion implementation computes a real **multivariate Matrix Profile** and saves the motif/discord figure shown here.
- Motifs and discords are often easier to explain to a room than a latent embedding.
- The teaching comparison is now honest and direct: **DTW** when you have a known healthy template, **Matrix Profile** when you want self-similarity mining.
- If the real-data panel is much larger, keep the same interpretation but switch the implementation to `stumpy.mstump` for scale.

Fusion must be calibrated

- Bad claim: *“Fusion always beats every single branch.”*
- Honest claim: *“Calibrated fusion often improves review-list quality and robustness — when component scales are aligned to representative feedback.”*

Design options worth naming: - **Weighted sum / max** after normalization — simple score-level baselines. - **Rank fusion** — stable when component score scales drift. - **Validation-calibrated logistic fusion** — the main training example; teachable and operationally honest.

Reporting should expose the **dominant branch** and **most suspicious channels**, not only a final scalar.

Calibration formulas

Each component score is standardized from healthy training windows:

$$Z_{i,b} = \frac{S_{i,b} - \mu_b^{\text{train-normal}}}{\sigma_b^{\text{train-normal}} + \epsilon}$$

Simple score-level baselines:

$$s_i^{\text{mean}} = \frac{1}{B} \sum_b Z_{i,b}, \quad s_i^{\text{max}} = \max_b Z_{i,b}$$

Interpretation: standardization makes branch scores comparable enough to fuse. Without it, the largest-scale detector wins by accident.

Logistic fusion — formula and reading

The training example fits the fusion layer on validation labels:

$$p_i = \sigma \left(\beta_0 + \sum_b \beta_b z_{i,b} \right), \quad \sigma(u) = \frac{1}{1 + e^{-u}}$$

- Positive β_b : higher branch score increases alert odds after other branches are considered.
- Near-zero β_b : little extra validation signal.
- Negative β_b : the branch was inversely useful on this validation slice; inspect before generalizing.

How to interpret branch disagreement

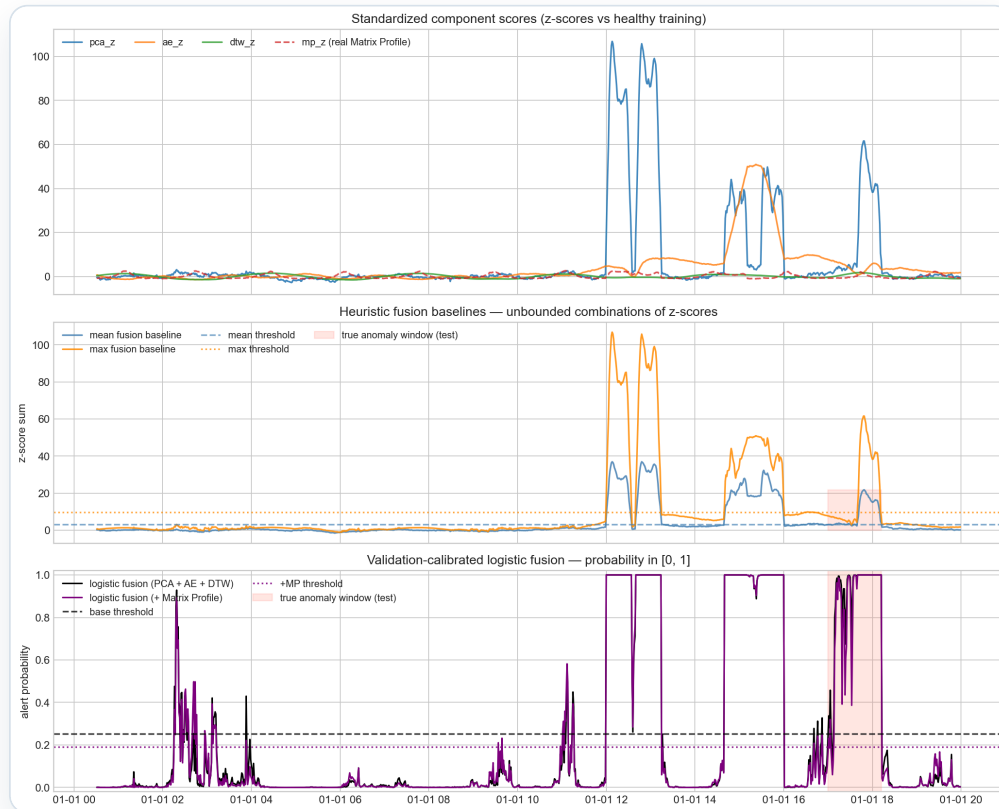
If PCA is high but DTW is quiet: - likely a magnitude / covariance event

If DTW is high but PCA is quiet: - likely a shape change with modest absolute magnitude

If AE is high while the others are mixed: - inspect whether the anomaly is nonlinear or channel-interaction heavy

If all branches fire: - confidence rises, but you still need to inspect **which channels changed**

Demo result from the training example



The teaching objective is not “black-box fusion.” It is **component diversity plus explicit calibration.**

What the example showed

Per-branch and fused metrics on the reference demo seed:

Branch / fusion	Precision	Recall	F1
PCA	1.00	0.88	0.93
AE	0.43	0.64	0.52
DTW	1.00	0.30	0.46
Matrix Profile	0.00	0.00	0.00
Fused (PCA+AE+DTW)	0.96	0.95	0.95
Fused (+ Matrix Profile)	0.96	0.92	0.94

Headline: calibrated **PCA + AE + DTW** fusion beat the **+ Matrix Profile** variant on held-out F1 for this reference seed.

How to read the result

Calibrated logistic fusion weights:

- Base fusion: $pca_z + 2.41$, $ae_z + 0.01$, $dtw_z - 0.61$
- +MP fusion: $pca_z + 2.31$, $mp_z + 0.50$, $ae_z + 0.04$, $dtw_z - 0.90$

Teaching reading: - The real Matrix Profile branch ran honestly and produced motif/discord review artifacts, but it **did not improve held-out F1 on this reference seed**. - That is exactly the professional lesson: an extra branch can be explanatory or useful without guaranteeing additive lift in fusion. - Calibration still mattered most: the strongest **single branch** remained **PCA**, while DTW was down-weighted and AE stayed near-zero.

Evaluation formulas used in the example

Point-level metrics on the held-out test slice:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Interpretation: precision protects operator attention; recall protects missed incidents. F1 is a convenient teaching summary, but it does not capture alert latency or event overlap quality.

Honest evaluation caveat

These numbers are point-style Precision/Recall/F1 on a single labeled seed. Treat them as a teaching artifact, not a benchmark claim. The items below matter in practice, but they are **not computed in this demo run**:

- **Review-list quality / Precision@k** is closer to how operators actually consume alerts than a single global F1.
- **Range-aware / event-level scoring** (e.g. Tatbul range metrics, NAB protocol) handles anomalous segments; point-adjust scoring is known to inflate results.
- **Detection latency** matters as much as accuracy: a correct alert that arrives late can be operationally useless.
- **Benchmark skepticism** still applies: common protocols and datasets can exaggerate apparent gains, so keep humility when reporting any fusion win.

How to facilitate the exercise

Companion implementation: multivariate panel walkthrough.

Recommended facilitation checkpoints: - inspect the raw panel before scoring, - show the lag-aware correlation matrix, - explain branch-specific scores separately, - contrast **DTW** (known template) with the runnable **Matrix Profile** branch (self-similarity mining), - calibrate fusion on validation labels, - review the top anomalous windows and the top contributing branch.

Classroom discussion prompts

- Why keep PCA once we already have an AE?
- What kind of anomaly would a template branch catch first?
- What would you do if you had no validation labels?
- Is the fused score helping because it is “smarter,” or because it is better calibrated?

Common pitfalls

- averaging scores without thinking about calibration,
- treating fused scores as self-explanatory,
- not checking which component dominated the alert,
- assuming Matrix Profile must improve fusion just because it adds another branch,
- confusing “multivariate” with “throw every metric into one model.”

Bridge to root-cause investigation

By this point, learners should be able to save: - top anomalous windows, - dominant components, - suspicious channels, - and candidate false positives to revisit with context and directional reasoning.

That bridge matters because the next step is moving from “**this window is suspicious**” to “**what likely changed first, under what context?**”.